

Taking into account that $\text{Var}(t, \Delta t) = \frac{s(t+\Delta t) - s(t)}{\Delta t}$

we rewrite expression (2) in the following form

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{s(t+\Delta t) - s(t)}{\Delta t} \quad (3)$$

As a result, we can formulate the following definition of the instantaneous velocity.

Definition: The instantaneous velocity at a moment of time t is the limit of a sequence of average velocities over time intervals from t to $t+\Delta t$ for $\Delta t \rightarrow 0$.

Reader: Now I realize that instead of talking about a sufficiently small time interval Δt (I am referring to our talk about the ratio $\frac{\Delta s}{\Delta t}$ at the beginning of the dialogue), the argument should have been based on the **limit transition** for $\Delta t \rightarrow 0$. In other words, the instantaneous velocity is not $\frac{\Delta s}{\Delta t}$ at a sufficiently small Δt but $\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$.

Author: Exactly. The definition formulated above for the instantaneous velocity not only exposes the gist of the concept but gives a rule for its calculation, provided that an analytical expression for $s(t)$ is known. Let us make such a calculation assuming that $s(t)$ is given by expression (1).

Reader: We should substitute (1) for (3). This gives

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\frac{a(t+\Delta t)^2}{2} - \frac{at^2}{2}}{\Delta t}$$

Author: Go ahead. Remove the parentheses.

Reader: This will give $v(t) = \lim_{\Delta t \rightarrow 0} \frac{a(t^2 + 2t\Delta t + \Delta t^2 - t^2)}{2\Delta t} = \lim_{\Delta t \rightarrow 0} (at + \frac{\Delta t}{2})$

$$= at.$$

We have arrived at a familiar formula for the velocity of uniformly accelerated motion with zero initial velocity: $v(t) = at$ (4)